Screenshot 1A: Output of `./test_objects` showing all tests passing.

```
❯ cd PES1UG24CS526-pes-vcs
❯ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_write':
object.c:147:47: warning: '.tmp' directive output may be truncated writing 4 by
tes into a region of size between 1 and 512 [-Wformat-truncation=]
  147 |     snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
      |                                                 ^~~~
In file included from /usr/include/stdio.h:980,
                 from pes.h:8,
                 from object.c:11:
In function 'snprintf',
    inlined from 'object_write' at object.c:147:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_
chk' output between 5 and 516 bytes into a destination of size 512
   54 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   55 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   56 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4
b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bb
e9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
~/os-orange/PES1UG24CS526-pes-vcs  main !1                    base Py
❯
VibhaVasisht (16 minutes ago)
```

Screenshot 1B: `find .pes/objects -type f` showing the sharded directory structure.

```
❯ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
~/os-orange/PES1UG24CS526-pes-vcs  main !1                    base Py
❯
```

Screenshot 2A: Output of `./test_tree` showing all tests passing.

```
❯ make test_tree
./test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_write':
object.c:147:47: warning: '.tmp' directive output may be truncated writing 4 by
tes into a region of size between 1 and 512 [-Wformat-truncation=]
  147 |     snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
      |                                                ^~~~
In file included from /usr/include/stdio.h:980,
                 from pes.h:8,
                 from object.c:11:
In function 'snprintf',
    inlined from 'object_write' at object.c:147:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_
chk' output between 5 and 516 bytes into a destination of size 512
   54 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   55 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   56 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
~/os-orange/PES1UG24CS526-pes-vcs main !4                    base Py
❯
```

Screenshot 2B: Pick a tree object from `find .pes/objects -type f` and run `xxd .pes/objects/XX/YYY... | head -20` to show the raw binary format.

```
❯ xxd .pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e5
66fcf | head -20
00000000: 626c 6f62 2031 3500 5465 7374 2069 6e74  blob 15.Test int
00000010: 6567 7269 5879 0a                        egriXy.
~/os-orange/PES1UG24CS526-pes-vcs main !4                    base Py
❯
```

Screenshot 3A: Run `./pes init`, `./pes add file1.txt file2.txt`, `./pes status` — show the output.

```
❯ make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  Makefile
  untracked:  index.c
  untracked:  test_objects
  untracked:  object.c
  untracked:  tree.c
  untracked:  .gitignore
  untracked:  README.md
  untracked:  index.h
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  commit.c
  untracked:  test_objects.c
  untracked:  commit.h
  untracked:  tree.h
  untracked:  pes.h
  untracked:  pes.c

~/os-orange/PES1UG24CS526-pes-vcs  main !4 ?2                base Py
❯
```

`cat .pes/index` showing the text-format index with your entries.

```
❯ cat .pes/index
100644 626c6f6220360068656c6c6f0a0000000000000000000000000000000000000000 1776682
350 6 file1.txt
100644 626c6f62203600776f726c640a0000000000000000000000000000000000000000 1776682
350 6 file2.txt
~/os-orange/PES1UG24CS526-pes-vcs  main !4 ?2                base Py
❯
```

Screenshot 4A: Output of `./pes log` showing three commits with hashes, authors, timestamps, and messages.

```
❯ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: 181f5b0e5626... Initial commit
Committed: 632564372d01... Add world
Committed: 612a63352b0d... Add farewell
commit 612a63352b0d0e1a4f4e301b4b31626b0e3e557a0f5400183e43517554594624
Author: PES User <pes@localhost>
Date:   1776682943

    Add farewelldd f

commit 632564372d01091a4c1e331d4e3167670c6f522c0d0d020b34495d3236612825
Author: PES User <pes@localhost>
Date:   1776682943

    Add worldelldd f

commit 181f5b0e5626257330783278100d45483b0a1030595a4210721b0e0b42222c28
Author: PES User <pes@localhost>
Date:   1776682943

    Initial commit

~/os-orange/PES1UG24CS526-pes-vcs  main !1 ?2                base Py
❯
```

Screenshot 4B: `find .pes -type f | sort` showing object store growth after three commits.

```
❯ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/18/1f5b0e5626257330783278100d45483b0a1030595a4210721b0e0b42222c28
.pes/objects/52/6e6e28772b6f3478166c0d0e4f54205a752d4245306f72275c0c084b583423
.pes/objects/52/6e6e28772b6f3478166c0d0e4f54205d47656f4c306c172d0b637a273c3e23
.pes/objects/61/2a63352b0d0e1a4f4e301b4b31626b0e3e557a0f5400183e43517554594624
.pes/objects/62/6c6f622031320048656c6c6f0a576f726c640a0000000000000000000000000
.pes/objects/62/6c6f622031350054657374206 96e746567726974790a000000000000000000
.pes/objects/62/6c6f622031350068656c6c6f20776f726c6420313233000000000000000000
.pes/objects/62/6c6f622031360048656c6c6f2c205045532d564353210a0000000000000000
.pes/objects/62/6c6f622031380044757 06c696361746520636f6e74656e740a000000000000
.pes/objects/62/6c6f6220360048656c6c6f0a0000000000000000000000000000000000000
.pes/objects/62/6c6f6220360068656c6c6f0a0000000000000000000000000000000000000
.pes/objects/62/6c6f62203600776f726c640a0000000000000000000000000000000000000
.pes/objects/62/6c6f62203800476f6f646279650a0000000000000000000000000000000000
.pes/objects/63/2564372d01091a4c1e331d4e3167670c6f522c0d0d020b34495d3236612825
.pes/refs/heads/main
  ~/os-orange/PES1UG24CS526-pes-vcs  main !1 ?2                    base Py
❯
```

Screenshot 4C: `cat .pes/refs/heads/main` and `cat .pes/HEAD` showing the reference chain.

```
❯ cat .pes/refs/heads/main
612a63352b0d0e1a4f4e301b4b31626b0e3e557a0f5400183e43517554594624
  ~/os-orange/PES1UG24CS526-pes-vcs  main !1 ?2                    base Py
❯
```

```
❯ cat .pes/HEAD
ref: refs/heads/main
  ~/os-orange/PES1UG24CS526-pes-vcs  main !1 ?2                    base Py
❯
```

Make test integration (final)

```
❯ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 1e4358065677... Initial commit

Log after first commit:
commit 1e43580656772b27342c3227160c46473b5716365a0947107e17085c45272a2d
Author: PES User <pes@localhost>
Date:   1776683157

    Initial committial commiQ


--- Second Commit ---
Committed: 150766672c51... Update file.txt

--- Third Commit ---
Committed: 682161342b06... Add farewell

--- Full History ---
commit 682161342b0609454f42314a48386e310a68502d090d04143d15557b540a4127
Author: PES User <pes@localhost>
Date:   1776683157
```

```
--- Full History ---
commit 682161342b0609454f42314a48386e310a68502d090d04143d15557b540a4127
Author: PES User <pes@localhost>
Date:   1776683157

    Add farewelldd f

commit 150766672c510b484d4837174e346a310e38553e1d0c44043a1d556b5d580605
Author: PES User <pes@localhost>
Date:   1776683157

    Update file.txtf

commit 1e43580656772b27342c3227160c46473b5716365a0947107e17085c45272a2d
Author: PES User <pes@localhost>
Date:   1776683157

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
682161342b0609454f42314a48386e310a68502d090d04143d15557b540a4127

--- Object Store ---
Objects created:
10
.pes/objects/15/0766672c510b484d4837174e346a310e38553e1d0c44043a1d556b5d580605
.pes/objects/1c/39334f41436a433c1f03002832244871707b32336362546234455e045f785c
.pes/objects/1e/43580656772b27342c3227160c46473b5716365a0947107e17085c45272a2d
.pes/objects/62/6c6f62203130007665727369 6f6e20310a0000000000000000000000000000
.pes/objects/62/6c6f622031320068656c6c6f20776f726c640a0000000000000000000000000
.pes/objects/62/6c6f62203230007665727369 6f6e20310a76657273696f6e20320a00000000
.pes/objects/62/6c6f62203800676f6f646279650a0000000000000000000000000000000000
.pes/objects/67/7b7b796722215400582a3c2467783b07243248181754773c2e33301e061068
.pes/objects/67/7b7b796722215400585c595614115469040042181754773c2e33301e051068
.pes/objects/68/2161342b0609454f42314a48386e310a68502d090d04143d15557b540a4127

=== All integration tests completed ===
~/os-orange/PES1UG24CS526-pes-vcs main !1 ?2                    base Py
❯
```

Phase 5: Analysis Questions

Q5.1

A branch is a file storing a commit hash. To implement checkout:

- Update HEAD to point to the branch
- Read the commit tree of that branch
- Replace working directory files with tree contents

Complexity arises due to handling uncommitted changes and file conflicts.

Q5.2

Compare working directory files with index entries:

- If a file differs from the index and also differs in the target branch → conflict
- Reject checkout if mismatch exists

Q5.3

Detached HEAD stores a commit directly:

- New commits are not attached to any branch
- Can recover using the commit hash and creating a new branch

---

Phase 6: Garbage Collection

Q6.1

Use graph traversal (DFS/BFS):

- Start from all branch heads
- Mark reachable objects
- Delete unmarked objects

Use a hash set to track visited objects.

Q6.2

Garbage collection during commit can delete objects being written:

- Causes race conditions between commit and GC

Git avoids this using:

- Lock files
- Reference tracking
- Delayed garbage collection

Conclusion

Successfully implemented a Git-like version control system with object storage, tree structures, indexing, and commit history management.